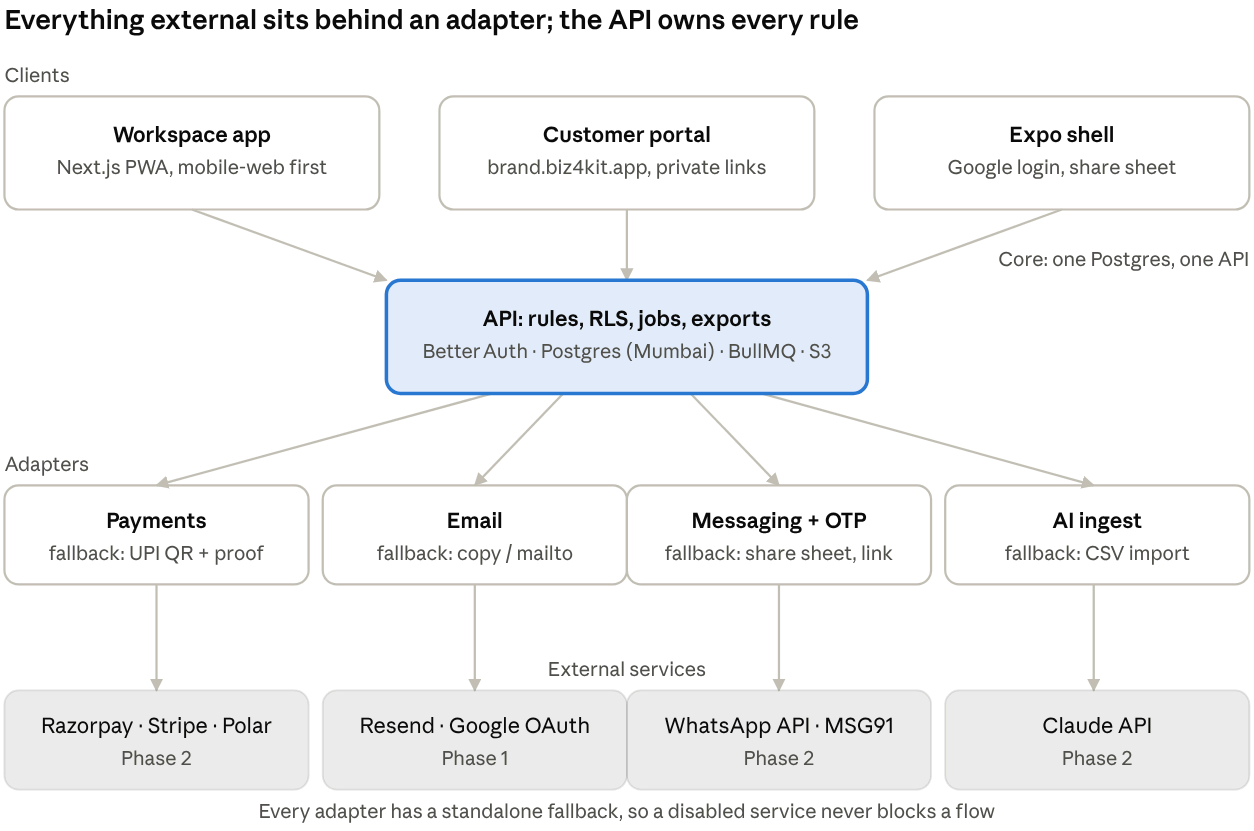
Architecture & contracts

Oct 18, 2026 · @Ravin

One Next.js app serves the workspace and the customer portal, one API owns every rule, one Postgres with row-level security holds every workspace, and everything external (payments, email, messaging, AI) sits behind an adapter with a standalone fallback. This tab is the spec AI agents build from; the Data model section of Design inputs defines the entities it refers to.

System architecture



system architecture · 3 clients, 1 core, 4 adapters, external services by phase

Clients never talk to an external service directly; every call goes through the API and then an adapter, so a missing or disabled connector always lands on the Phase 1 fallback. In Phase 1 only the email adapter and Google OAuth have a real provider behind them.

Services and modules

One deployable API, split into modules with an owner each; a module may only call another through its exported service, never its tables.

Module	Owns	Exposes	Owner (fill in)
`auth`	Better Auth config, sessions, invites, roles, re-auth	`requireUser`, `requireRole`, invite service	
`workspace`	Workspace, Member, brand theme generation, numbering settings	theme tokens, numbering allocator	
`catalog`	CatalogItem, Package, CSV import, public catalog read model	item and package services, public page data	
`leads`	Lead, portal request form, assignment	lead service, source tagging	
`documents`	Quote and Invoice blocks, versions, totals, GST, QuoteLine snapshots, render model shared by web and PDF	quote service, invoice service, `renderDocument`	
`payments`	Payment, proof files, approval queue, thresholds, receipts	payment service, approval service	
`bookings`	Booking, availability, conflicts, Job, checklists	booking and job services	
`customers`	Customer, timeline, My-orders tokens	customer service, timeline read model	
`messages`	Templates, rendering, share logging, in-app inbox, notification routing	`notify(event, record)`	
`exports`	PDF / DOCX / CSV jobs, storage, signed URLs	`export(record, format)`	

Module	Owns	Exposes	Owner (fill in)
`connectors`	Adapter registry, per-workspace settings, secrets, health	`getAdapter(kind, workspace)`	
`admin`	Super-admin console, abuse controls, audit views	admin service	
`jobs`	BullMQ queues: pdf, email, expiry, reminders (Phase 2), ingest (Phase 2)	queue names and payload schemas	

Shared packages: `db` (schema, migrations, RLS policies), `ui` (design system), `templates` (React templates for web and PDF), `connectors` (adapter interfaces and implementations).

API contract

Internal calls use tRPC with shared types; the customer portal and Phase 3 public API use REST on the same services. Conventions first, then the Phase 1 surface.

Conventions

- Base `/api/v1`; JSON; UTC timestamps in ISO 8601; money as integer paise with a `currency` field.
- Auth: session cookie for members (Better Auth); `Authorization: Bearer <portal token>` for customer links; every request resolves a `workspace_id` or is rejected before any query.
- Errors: `{ error: { code, message, field? } }` with codes `unauthenticated`, `forbidden`, `not_found`, `validation`, `conflict`, `rate_limited`, `state` (invalid transition); HTTP status matches.
- Pagination: cursor-based, `?cursor=&limit=` (max 100); list responses return `{ items, next_cursor }`.
- Idempotency: `Idempotency-Key` header required on create and on any money or send action; replays return the first result.
- Rate limits: portal endpoints 60 requests per minute per IP and per token; member endpoints 600 per minute per workspace.

- Versioning: additive changes only within v1; a breaking change is v2.

Phase 1 endpoints (members)

Method	Path	Does
POST	`/workspaces`	Create workspace (onboarding step 2)
PATCH	`/workspaces/:id/brand`	Logos, colours; returns generated theme tokens and contrast warnings
POST	`/workspaces/:id/invites` · `/invites/:token/accept`	Invite, accept; invite code variant `POST /invites/claim`
GET/POST/PATCH	`/catalog/items`, `/catalog/items/:id`	Items with variants and add-ons; `POST /catalog/import` for CSV with a mapping preview
GET/POST/PATCH	`/packages`, `/packages/:id`	Packages and tiers
GET/POST/PATCH	`/leads`, `/leads/:id` · `POST /leads/:id/assign`	Leads inbox
POST	`/quotes` (from lead or package) · `PATCH /quotes/:id` (draft blocks) · `POST /quotes/:id/share` · `POST /quotes/:id/revise`	Quote lifecycle; share returns the private link and the ready message
POST	`/invoices` (from quote, job or manual) · `POST /invoices/:id/share` (assigns the number) · `POST /invoices/:id/void`	Invoice lifecycle
POST	`/invoices/:id/payments` (amount, method, proof files) · `POST /payments/:id/approve` · `POST /payments/:id/reject`	Record and approve payments; approve also accepts a signed email token
GET/POST/PATCH	`/bookings`, `/bookings/:id` · `GET /availability`	Calendar

Method	Path	Does
GET/PATCH	`/jobs/:id` · `POST /jobs/:id/complete`	Jobs; complete creates the draft invoice
GET	`/customers`, `/customers/:id/timeline`	Customers
POST	`/records/:type/:id/share`	Logs a Message row and returns the rendered text for the chosen channel
POST	`/exports` · `GET /exports/:id`	Queue and fetch PDF / DOCX / CSV
GET	`/inbox` · `POST /inbox/:id/read`	In-app notifications
GET	`/reports/summary?from&to`	Reports v1

Phase 1 endpoints (customer portal, token auth)

Method	Path	Does
GET	`/p/:slug/catalog`	Public catalog read model (cacheable)
POST	`/p/:slug/requests`	Quote request → Lead (rate-limited, consent flag required)
GET	`/q/:token` · `POST /q/:token/accept`	Quote page and acceptance (name, phone, selected optional lines)
GET	`/book/:package/slots` · `POST /book/:package`	Booking picker
GET	`/i/:token` · `POST /i/:token/proof`	Invoice page, UPI deep link and QR, proof upload
GET	`/me/:token`	My orders

Webhooks (Phase 2, inbound): `/webhooks/:provider` with signature verification and an idempotency table; unmatched payment webhooks create a `pending_approval` Payment rather than failing silently.

Connector adapter interface

Four adapter kinds, one registry. A workspace enables zero or more providers per kind; the API asks the registry for the best available adapter and always receives one, because the standalone fallback is itself an adapter.

Kind	Interface (TypeScript sketch)	Fallback adapter (Phase 1)	Providers (Phase 2)
Payments	`createPaymentRequest(invoice) → { kind: 'link' | 'static', url?, upi?, bank?, qr }`, `verifyWebhook(req)`, `reconcile(event) → Payment`, `refund?()`	`static-upi`: returns the UPI deep link, QR and bank block; reconciliation is the approval queue	`razorpay`, `stripe`, `polar`
Email	`send(template, to, data) → { id }`, `status(id)`	`mailto`: returns a ready message and logs it as shared; used only if Resend is down	`resend` (Phase 1 default), SES later
Messaging	`send(channel, template, to, data)`, `status(id)`, `inbound(req)`	`share-sheet`: returns the rendered text and a `wa.me` / `sms:` link	`whatsapp-business`, `msg91`
OTP	`start(phone) → { challenge }`, `verify(challenge, code)`	`link-possession`: the private link is the proof; no code	`msg91`, `twilio`
Ingest	`structure(file | text, target) → { draft, confidence }`	`csv`: column mapping only	`claude`

Rules

- Adapters never touch the database; they return typed results and the owning module writes.
- Each provider has a `health()` check and a settings schema; the settings UI is generated from the schema, with secrets write-only.

- Selection order per kind is a workspace setting with a recommended default (Razorpay in India); if the chosen provider fails health, the next enabled one is used, then the fallback, and the owner is told in the inbox.
- Every outbound call is logged with provider, latency and a scrubbed payload; every inbound webhook is stored raw before processing.
- Contract tests run against each provider's sandbox nightly and against a recorded fixture on every PR.

Analytics event plan

Events are named `object_verb`, carry `workspace_id`, `actor` (owner / staff / customer / system) and `source` where known, never a name, phone or email. One event per row below is enough to compute every GTM test.

Event	Fired when	Properties	Feeds
`workspace_created`	Onboarding step 2	vertical, channel tag from sign-up link	G1, G7
`catalog_shared`	Owner copies the link, downloads the QR or shares the price-list message	method	G1
`lead_created`	Portal request or manual entry	source (qr / web / instagram / referral / manual)	G2, G8
`quote_shared`	First share of a quote	minutes_since_lead, from_package	G2
`quote_viewed`	First portal open, bots excluded	hours_since_share	G3
`quote_accepted`	Customer accepts	version, hours_since_share, optional_lines_selected	G3
`invoice_shared`	Number assigned and shared	from (quote / job / manual), total_bucket	G4
`payment_recorded`	Proof uploaded or cash recorded	method, has_proof	G4

Event	Fired when	Properties	Feeds
`payment_approved`	Owner or Manager approves	hours_since_recorded, via (app / email)	G4
`invoice_paid`	Invoice turns Paid	days_since_share	G4
`booking_confirmed`, `job_completed`	As named	staff_count	usage
`member_active_day`	First action of a member each day	role	G5
`commitment_asked`, `commitment_answered`	Day-20 ask logged by the success buddy	answer (yes / no / later)	G6
`signup_attributed`	Workspace created from a tagged link or the free-plan badge	channel	G7
`template_shared`	Any ready message copied or shared	event number, channel	messages
`export_created`	PDF / DOCX / CSV	format, record type	usage

Dashboards: one per GTM test with its pass line drawn; a daily snapshot posts to Slack #pilot. Event names are frozen once the pilot starts; new events are added, old ones never renamed.

Acceptance criteria (Phase 1)

Each line becomes a test in the package's first commit and a checkbox in its Linear issue. The WBS "Done when" column is the summary; this is the contract.

WBS	Given / when / then
1.2	Given a member of workspace A, when they query any table with B's ids, then zero rows and no error leak; given a migration, when run twice, then it is a no-op
1.3	Given a Google account, when it signs in on web and on the Expo shell, then one user, two sessions, both listed and individually revocable; given an invite to `a@x`, when `b@y` signs in, then the invite is visible to the owner, editable, and claimable with the invite code; given a billing or delete action, when the session is older than 15 minutes, then re-auth is required
1.4	Given two brand colours, when saved, then tokens exist for light and dark and every text-on-primary pair is ≥ 4.5:1 or the owner is warned and the colour adjusted
1.6	Given each notification event, when it fires, then exactly one email is sent to the right person with a working single-use link; given a bounce, then the member sees a warning in settings
1.7	Given airplane mode, when staff open today's bookings and jobs and draft a quote, then data is shown from cache, the draft is kept, and syncs on reconnect with a visible conflict banner if the server changed
2.1	Given a CSV with Vyapar headers, when imported, then columns are mapped with a preview and unmapped columns are listed; given a hidden-price item, when the portal renders, then price is replaced by "ask for a quote"; given a WhatsApp preview bot, when it fetches the catalog, then no analytics event fires

WBS	Given / when / then
2.3	Given a quote edited on desktop and the same quote opened on a phone, when lines change on either, then totals match to the paisa and the PDF matches the web render pixel-for-pixel in the visual test
2.4	Given a shared quote, when the customer opens a forwarded link, then they can accept with name and phone and the acceptance stores version, IP and time; given an expired quote, when accept is attempted, then `state` error; given a revise, then the number is unchanged and version + 1
2.5	Given 100 invoices shared in parallel, then numbers are gap-free and unique; given a draft, then no number; given customer state ≠ workspace state, then IGST; given a void, then the number is retained and the invoice is excluded from totals
2.6	Given a proof upload by staff, when the owner taps the email link, then the payment is approved once and the link is dead afterwards; given a Manager and an amount above the threshold, then approve is forbidden; given approval, then the invoice turns Paid and the receipt message is offered
2.7	Given every template and a sample record, when rendered, then no unresolved `{variable}` remains and WhatsApp length ⩽ 60 words; given a share, then a Message row exists on the record
2.8	Given any quote or invoice, when exported, then PDF, DOCX and CSV are produced within 3 s (p95) and the DOCX opens in Word and Google Docs with the brand header intact
2.9	Given a new lead, when created, then the assignee gets one inbox item and one email; given the same event twice (idempotency replay), then still one

WBS	Given / when / then
3.1–3.3	Given overlapping bookings for one member, then the second is refused with a conflict; given a job marked done with an incomplete required checklist, then refused; given done, then a draft invoice exists linked to the job
3.4	Given a customer's My-orders link, then every quote, booking, invoice and receipt of that customer and no other customer's
3.5	Given each role, when each endpoint is called, then the permission matrix holds (Staff cannot read reports or approve; Accountant is read-only); every mutation writes an ActivityLog row
3.7–3.8	Given a slug containing a reserved or bank-like word, then creation is refused; given an abuse report, then the workspace is flagged in the console within one request; given admin impersonation, then a consent log entry exists
4.1	Given the full suite, then it runs green in under 5 minutes; given Lighthouse on the four portal pages, then LCP ⩽ 2.5 s under the NFR profile; given axe, then zero serious violations
4.3	Given the Expo shell, when installed on three Android phones and left a week, then the session is still valid and the share sheet opens WhatsApp with the message prefilled

Test plan and device matrix

Scope by day: unit and integration from day 1; end-to-end on previews from day 3; non-functional and security sweeps on days 8–9; a real-owner session on day 7 and day 10.

Device matrix (buy or borrow these before day 3)

Device	Why	Browser	Flows
Mid-range Android, 4 GB RAM, Android 12–14 (Moto G / Redmi class)	The staff and customer phone	Chrome, WhatsApp in-app browser, Instagram in-app browser	All four flows, offline test, share sheet
Low-end Android, 2–3 GB RAM, Android 10–11	Worst case customer	Chrome	Catalog, quote accept, pay page
iPhone (two versions back)	Owners and agency clients	Safari	Portal flows, PDF download
Laptop, Windows + Chrome; Mac + Safari	Owners in the evening	Chrome, Safari, Edge	Workspace, doc editor, exports
Printer	Invoices and QR cards are printed	—	PDF print at A4, QR scannable from paper

Network profiles: Lighthouse Slow 4G for budgets; a throttled 3G run once for the pay page.

Data: generated demo workspaces for each vertical in preview and staging; one consented real pilot owner's data in production on days 7 and 10; never real data in lower environments.

Exit criteria: all acceptance tests green; zero open severity-1 or severity-2 bugs; device matrix passed by a human on days 9–10 with evidence in Linear; NFR budgets met; the go / no-go checklist in the Workflow tab.

Vendor and account checklist

Start everything in the first column on day 1; lead times are from general experience and vary, so confirm each with the provider.

Account	Needed for	Phase	Typical lead time	Start	Owner (fill in)
Domain, DNS, TLS	Everything	1	Hours	Day 1	

Account	Needed for	Phase	Typical lead time	Start	Owner (fill in)
Google Cloud project, OAuth consent screen (external, verified)	Google sign-in	1	Hours; verification days to weeks if sensitive scopes are requested (avoid them)	Day 1	
Resend account, sending domain with SPF / DKIM / DMARC	Email	1	Hours; warm-up over a week for volume	Day 1	
Cloud hosting (Vercel / Fly.io), managed Postgres and S3-compatible storage in Mumbai	Infra	1	Hours	Day 1	
Error tracking, analytics, uptime, status page	Observability	1	Hours	Day 2	
GitHub organisation, Linear workspace, Slack workspace	Workflow	1	Hours	Day 1	
WhatsApp Business number for support	Human support	1	Days (number + app)	Day 2	
Meta Business verification and WhatsApp Business API access, template approval	WhatsApp automation	2	Days to several weeks	Day 1	
SMS provider (MSG91 / Twilio) and Indian sender / template registration	OTP, SMS	2	Days to weeks (regulatory registration)	Day 1	
Razorpay account and KYC	Payments, plan billing	2	Days; depends on documents	Day 1	
Stripe account (confirm Indian eligibility first)	International payments	2	Unknown; may be restricted	Day 1	
Polar account (confirm service-business eligibility first)	International billing as merchant of record	2	Days	Day 1	
Claude API billing account and usage caps	AI ingest	2	Hours	Day 11	
Apple Developer and Google Play accounts	Native app	3	Days; Apple requires a legal entity	Week 4	

Account	Needed for	Phase	Typical lead time	Start	Owner (fill in)
Company registration, GST, bank account	Charging money	before Phase 2 billing	Weeks	Now	
Trademark search and filing for the product name	Brand	before pilot	Search: days; registration: months	Now	

Each account is created with a shared company email, MFA on, and credentials in the secret store; no account lives under a personal login.